

Techno-economic analysis and dynamic power simulation of a hybrid solar-wind-battery-flywheel system for off-grid power supply in remote areas in Kenya

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ABSTRACT

Sub-Saharan Africa (SSA) has the lowest energy access rates globally. The need for transformative energy sources ranging from solar off-grid and mini-grid solutions to hybrid micro-grid power systems has rapidly grown to deliver clean energy admittance. This research proposes a hybrid photovoltaic-wind turbine power system coupled to a hybridized storage system composed of a Lithium-Ion battery and a flywheel storage system which ensures reliability for off-grid electrification for rural and less accessible remote areas of Makueni County in Kenya. The optimal size of the proposed Hybrid Renewable Energy System (HRES) is estimated, using a multi-objective optimization modeling approach, taking into account the Levelized Cost of Electricity (LCOE) and reliability (energy index of self-reliance (EISR)) of the system as the main objective functions, using epsilon (ϵ)-constraint technique as the solving approach. A resultant Pareto front is analyzed to obtain the best compromise for COE at 0.519 USD/kWh and reliability indicator, energy index of self-reliance (EISR) at 0.997. The optimal size of the HRES was realized at 26 PV panels (330 W) and 3 wind turbines (1 kW) which satisfies the annual local load requirement of 37.94MWh. Next, the hourly performance of the proposed HRES, under different operating conditions, is evaluated, using a dynamic power control simulation model developed in Matlab-Simulink. The results revealed the proposed off-grid HRES is highly reliable, meeting the load demand sufficiently in all meteorological conditions experienced in the area of study. Furthermore, adopting a hybrid energy storage system (HESS) realized an annual potential of 858kWh storage capacity gain in the battery when coupled with the flywheel storage system.

Introduction:

Background:

According to the world bank and the International Renewable Energy Agency (IRENA) tracking sustainable development goal 7 (SDG7) report for 2022 on energy progress, over 600 million people have no access to electricity, while 2.4 billion people are heavily dependent on unhealthy and environmentally harmful fuels for cooking and lighting [1]. Access to affordable, reliable, sustainable, and modern energy will be a dream far from reality for around 620 million people by 2030. More than 50 % of the SSA population had no access to electricity in 2020 [1]. Kenya has 71.4 % of its population access to the national electrical grid network. Alarmingly, 80 % of Kenyans in rural households use kerosene

for lighting [2]. This highlights the necessity for rapid growth in electricity generation in SSA driven by increased demand due to the rise in population and development in economic sectors. In addition, intensified fears of environmental impacts due to the high reliability of fossil fuels in urban and remote locations require imperative solutions [3]. Renewable energy generation and clean energy technologies are active character players in future renewable energy transitions [4]. Integrating renewable energy sources and power storage is advantageous to creating affordable, reliable, and efficient power supply system [5]. A hybrid system bearing a combination of solar and wind sources optimally attenuates substantial power fluctuations due to climate and meteorological conditions that result in high levels of the unreliability of RES power systems [6]. Off-grid microgrid hybrid renewable energy systems (HRES) combined with hybrid storage systems provide an accurate solution to power shortages in remote areas of the globe [7]. This can be

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Nomenclature

<i>HRES</i>	Hybrid Renewable Energy System
<i>COE</i>	Cost of Energy (USD/kWh)
<i>EISR</i>	Energy Index of Self Reliance
<i>SSA</i>	Sub-Saharan Africa
<i>SDG</i>	Sustainable Development Goals
<i>GAMS</i>	General Algebraic Modelling System
<i>FESS</i>	Flywheel Energy Storage System
<i>BESS</i>	Battery Energy Storage System
<i>HESS</i>	Hybrid Energy Storage System
<i>EPRA</i>	Energy and Petroleum Regulatory Authority
<i>IRENA</i>	International Renewable Energy Agency
<i>DC</i>	Direct Current
CO_2	Carbon dioxide
P_{Gen}	Power generated (W)
<i>UME</i>	Unmet Energy
<i>RES</i>	Renewable Energy Sources
<i>MOPSO</i>	Multi-objective Particle Swarm Optimization
<i>LPSP</i>	Loss of Power Supply Probability
<i>DG</i>	Diesel/fossil Generators
<i>AC</i>	Alternating Current
<i>PMSG</i>	Permanent Magnet Synchronous Generator
<i>PMSM</i>	Permanent Magnet Synchronous Machine
<i>SVPWM</i>	Space Vector Pulse Width Modulation
<i>MPPT</i>	Maximum Power Point Tracking
<i>P&O</i>	Perturb & Observe
<i>LCOE</i>	Levelized Cost of Electricity (USD/kWh)
<i>SOC</i>	State of Charge (%)
<i>DOD</i>	Depth of discharge (%)
ε	Random value, epsilon
PI Controller	Proportional Integral Controller
OMC_i	Annual Operation and Maintenance Cost
RC_i	Replacement Cost (USD)
f_1	Objective function for COE
f_2	Objective function for UME
r	Discount Rate (%)
y	Total component Lifespan (Years)
i	Number of Component Type
<i>PBch</i>	Battery Power charge (W)
<i>PBdch</i>	Battery power discharge (W)
<i>PFWch</i>	Flywheel power charge (W)
<i>PFWdch</i>	Flywheel power discharge (W)
E_{bat}	Energy storage by battery (Wh)
E_{gen}	Electrical energy generated (Wh)
E_L	Electrical load (W)
I_{bat}	Battery current (A)
V_{bat}	Battery voltage (V)
T_c	Solar cell temperature (K)
T_a	Ambient temperature (K)
<i>STC</i>	Standard test conditions
G	Insolation at T_a (W/m^2)
$G_{(\text{STC})}$	Insolation at STC (W/m^2)
P_{pv}	PV power output (W)
$I_{m(\text{STC})}$	Maximum Current at STC (A)
I_{sc}	Short circuit current (A)

$I_{\text{sc}(\text{STC})}$	I_{sc} at STC (A)
N_m	Number of PV modules
R_s	Series resistance (Ω)
V_{oc}	Open circuit voltage (V)
$V_{m(\text{STC})}$	Maximum voltage at STC (V)
$\mu_{V_{oc}}$	Voltage temperature coefficient ($\%/^{\circ}\text{C}$)
$\mu_{I_{sc}}$	Current temperature coefficient ($\%/^{\circ}\text{C}$)
$T_{c(\text{STC})}$	Cell Temperature at STC (K)
R_p	Shunt resistance (Ω)
V_{th}	Thermal Voltage (V)
σ_{oc}	PV empirical adjustment parameter
h_{hub}	Turbine height (m)
h_{ref}	Reference turbine height (m)
α	Power law coefficient
v_{ref}	Wind Speed at reference height (m/s)
P_{rated}	Rated turbine power (W)
v_{rated}	Rated turbine wind speed (m/s)
$v_{\text{cut_in}}$	Minimum turbine wind speed (m/s)
$v_{\text{cut_out}}$	Maximum turbine wind speed (m/s)
P_{WT}	Power generated by the wind turbine (W)
I	PV Output current (A)
I_o	Diode saturation current (A)
V	Terminal PV voltage (V)
V_{mp}	Voltage at Maximum Power Point (V)
V_{oc}	Open circuit voltage (V)
V_{sd}, V_{sq}	Stator circuit voltages (V)
I_{sd}, I_{sq}	Stator circuit currents (A)
d, q	Stator reference frames
ω_m	Rotational speed (rad/s)
Ψ_{sd}, Ψ_{sq}	Flux linkages (Wb)
L_{sd}, L_{sq}	Leakage inductances (H)
L_{md}, L_{mq}	Magnetizing inductances (H)
Ψ_m	Linkage flux (Wb)
J	Load combined inertia (Kg/m^2)
F	Combined viscous friction (Ns/m^2)
T_f	Shaft static friction
T_m	Mechanical torque (Nm)
θ	Angular position (rad)
E_{Fw}	Kinetic energy stored by the flywheel (J)
I	Flywheel moment of inertia (Kg m^2)
r_o, r_i	Flywheel radius (outer & inner) (m)
w_f	Flywheel angular velocity (rad/s)
η_{Bat}	Battery efficiency (%)
n_{cyc}	Number of battery cycles
Q	Battery rated capacity (Ah)
Q_{loss}	Percentage of battery capacity loss (%)
$Q_{\text{loss}}^{\text{Cal}}$	Percentage of battery calendar loss (%)
$Q_{\text{loss}}^{\text{Cyc}}$	Percentage of battery cycle loss (%)
T	Battery cell temperature (K)
R	The gas constant ($\text{Jmol}^{-1}\text{K}^{-1}$)
f	Battery pre-exponential factor ($1/\text{day}^{1/2}$)
E_a	Battery activation energy (kJmol^{-1})
I_{rate}	Battery charge/discharge rate (C-rate)

ascribed to a lack of access to the utility grid, poor resource distribution, and an inability to balance supply and demand factors [8]. The performance and reliability of HRES depend on the superlative synchronization and interaction between demand, generation output, efficiency aspects, storage, and the energy management strategy [9].

Literature survey:

Previous research has been carried out on optimal sizing of the HRES system as a fundamental approach in addressing renewable energy fluctuation issues occasioned by intermittency. Al-Sherei et al. [10] articulated improvement on the scale of hybrid wind-solar power system

Table 1

Overview of related research papers on optimal sizing of the off-grid HRES.

Ref	RES Type	Optimization Model	Results	Location
[19]	PV/WTS/DGS/Battery	Single objective cost minimization, using different Algorithms; IGWO, GWO, and SSA.	-The optimum LCOE is achieved at 0.215 \$/kWh for 260 W PV module, 30 kW wind turbine, 100 kW DG and 12v/50Ah battery bank. -Preference is given to DG sources due to low fuel cost. -SSA has acceptable performance stability among the optimization techniques.	Egypt
[18]	Wind/PV/DG/Battery Power system	Single objective cost minimization, using the PSO technique	- The optimal LCE was obtained at 0.003 €/kW for a 1 kW wind turbine and 230 W PV, suggesting a preference for using wind power against PV and DG. - The consideration of a smaller power demand does not effectively reflect changes in the cost of power and reliability, which may differ when a large residential power demand scenario is considered.	France
[20]	Wind/PV/battery	Multi-objective RESCA method to optimize the HRES system based on reliability and economic constraints	-The optimal size of the HRES system at minimum Annual system cost (ASC) represents 80.02 % renewable energy fraction (REF).	USA
[15]	Solar/ Wind/ DG, and PTES/ BESS storage	Optimal sizing with Cost minimization, using Mixed Integer Linear Programming	-Consideration of the flywheel storage system as complementary storage to BESS brings about additional benefits in capacity gain and frequency. -80.12 % in cost savings is achieved using pumped thermal energy storage (PTES) to BESS.	Italy
[16]	PV/Wind/DG/Battery storage	Optimal sizing with cost minimization, using a Multi-Objective Self-	-95.51 % in renewable power fraction at 0.083 \$/kWh and 0.219 of LPSP.	Saudi Arabia

Table 1 (continued)

Ref	RES Type	Optimization Model	Results	Location
[21]	Wind/ PV/ Battery storage	Adaptive Differential Evolution (MOSaDE) algorithm -Multi-objective optimization of costs and reliability, using a Non-dominated Sorting Genetic Algorithm (NSGA-II) - Chance Constrained Programming (CCP) method to cater to the RES uncertainties, using Monte Carlo simulation.	-The highest power contribution of power is from PV at 59 %. - low costs DPSP (blackout probability) tends to be 100 %. When DPSP is between 15 % – 35 %, at the optimum COE. -The optimal reliability index is 0.8.	UK
(Zhang et al., 2021)	PV/Wind/ Battery storage	Multi-objective particle swarm optimization (MOPSO) as a scheduling model for the minimization of Cost and maximize the reliability of power	- Maximum power reliability at the minimum value of LPSP. -The COE and LPSP show a Pareto-trade-off with optimal COE between LPSP values of 2 – 8 %. - Higher reliability by adopting a hybrid storage system.	Taiwan
[17]	Wind with HESS (Battery and Supercapacitor)	Frequency-based optimal sizing (PPSO-GA) and control mechanism on MATLAB Simulink	-Strong relationship between the cost of energy and battery cutoff frequency - Optimal cost is obtained at cutoff frequency of 1.623 MHz, taking into consideration the impact of charge/discharge efficiency of the battery on the cost per day. -A 59.62 % decrease in COE and 1.82 years. Increases battery lifetime	China

hesitancy by sinking costs while simultaneously ensuring the obligation of power supply dependency. They developed two techniques for ensuring system reliability based on energy storage and control strategy (Al-Sherei et al., 2021). Sharma et al. [11] developed the multi-objective optimization problem to minimize the operating cost of HRES and the use of non-renewable energy resources and fuel emissions. Their research compared single-objective and multi-objective optimization techniques in performing a trade-off [11]. Ning et al.(2021) developed a multi-objective optimal scheduling model using multi-objective particle swarm optimization (MOPSO) to assess the reversal relationship between LPSP and power supply reliability. Additionally, their research proposed that the contribution of Energy Storage Systems (ESSs) can improve reliability and enhance the penetration of distributed generation resources [12].

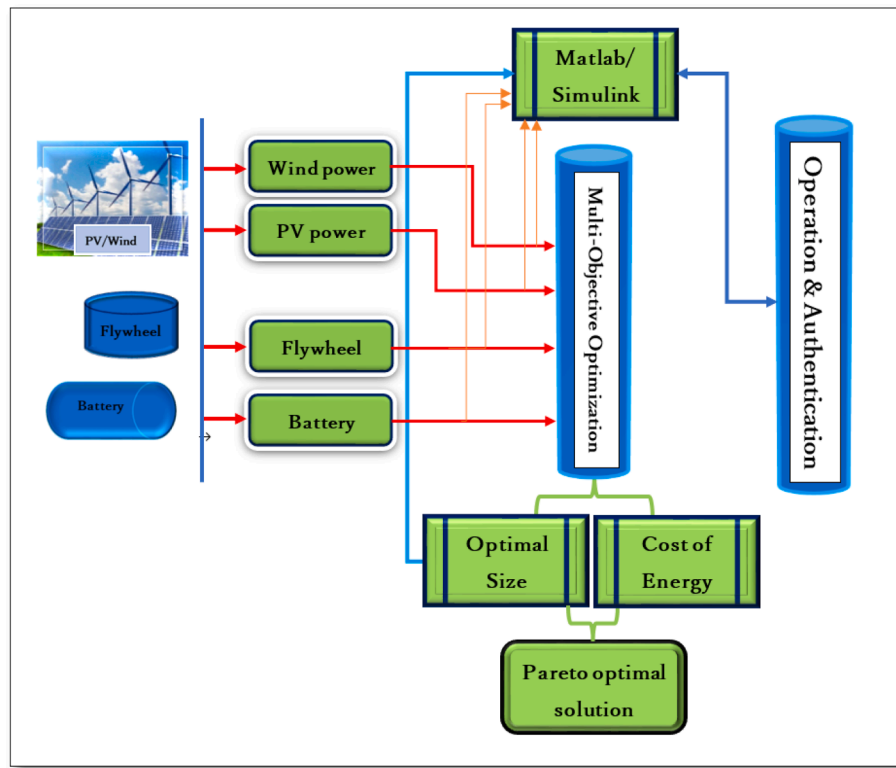


Fig. 1. Modeling framework in this study.

Various researchers considered dynamic modeling and simulation of HRES systems using Matlab-Simulink to evaluate system performance, control, and implement an efficient energy management strategy. Shaqour et al. [4] proposed the use of perturb and observe (P&O) control technique in maximum power point tracking (MPPT) coupled to a PI controller, which offers low cost and efficiency. Ni et al. (2021) demonstrated a finite control technique with a single-step prediction horizon for a simplified model predictive control (MPC) having efficient transient compared to P&O-based MPPT coupled to a PI controller, in which harmonic distortions are incurred. Benlahbib et al. (2020) explained an experimental study of a hybrid wind/solar/fuel cell/battery system. The study used MPPT-based single-input fuzzy logic (SIFL) controllers to optimize wind power, proposing a fuzzy logic controller (FLC) coupled to a programmable logic controller (PLC) unit to ensure effective energy management in the hybridized system.

Hybrid energy storage has been considered a strong alternative for achieving high performance of stand-alone or grid-connected microgrid renewable energy systems [13]. The benefit of power density and dynamic response is achieved through hybrid storage rather than single energy systems [14]. Babatunde et al. [14] demonstrated the significance of hybrid storage in renewable energy systems by underlining energy reliability improvement in fuel cell–battery systems as opposed to a single system of the fuel cell or battery alone. Acakpovi et al. (2020) described a detailed comparison of storage features during the simulation of an off-grid wind/hydrogen fuel-cell/battery system. This research deemed a stand-alone battery storage system economically viable; however, using a hybrid storage system with a hydrogen fuel cell is technically feasible due to increased autonomy and carbon credit gains. Roy et al. (2020) investigated the impact of depth of discharge (DoD) on the battery life in light of cycling and calendar aging effects, indicating the use of superconductors contributes to the prolonged battery service life.

Referring to the above-detailed discussion, it is obvious that efficient system performance and effective energy management are subsets of optimization, system sizing, dynamic system operation, and control

technique. Moreover, most previous studies only consider either output optimization and sizing or dynamic system operation and control without establishing a link between the two perspectives of the system modeling. Therefore, performing experimental examinations using appropriate engineering software makes it easier to authenticate the accuracy levels during optimization and component sizing.

Table 1 summarizes a comprehensive overview of related research papers based on their methodologies used in the optimal sizing of the HRES.

Research gap and originality highlights

From Table 1, several researchers have highlighted the methodology with little emphasis on the trade-off between reliability and COE, the use and significance of hybrid storage, and system operation control mechanisms. Contrary to other research works explained earlier, in which the HRES are optimized based on cost, this research dwells into multi-objective functions based on the comparison between cost and reliability, at the same time, authenticates the operation of the HRES using actual climatic data to illustrate the dynamic power control process used.

To this aim, first, the research establishes a convergence of optimum cost and energy reliability for the proposed HRES system, providing an appropriate size for PV panels, wind turbines, and energy storage based on the multi-objective ϵ -constraint mathematical programming. Secondly, it demonstrates a dynamic power operation using Matlab/Simulink simulation for the optimal sizes achieved in the optimization of the first part. A 72-hour simulation is carried out to illustrate the use of maximum power point tracking (MPPT)-controlled HRES system with a unique hybrid energy storage system (HESS) made up of flywheel and battery, taking into account the accurate authentication using hourly data for irradiance, temperature, wind speed and actual residential power demand collected in Makuani, which is the area of study to simulate the stand-alone system. This simulation further illustrates how the concept of HESS helps suppress frequency swings and prevent

voltage drops which are critical in microgrid steady frequency control. Fig. 1 represents the overall structure of the modeling framework developed in this study, considering size optimization and operation management of the proposed HRES.

The paper is structured as follows: Section 2 presents the hybrid renewable energy system configuration. The HRES size optimization modeling is explained in section 3. The section further describes the proposed multi-objective optimization method and the mathematical modeling for the power system components. Section 4 presents HRES dynamic operation and control modeling, describing the Matlab/Simulink simulation design for the system. Section 5 shows results and discussions for the multi-objective optimization and authentication on Matlab/Simulink of the HRES. Finally, the conclusion is presented in section 6.

HRES system configuration:

$$P_{Bdch(t)} + P_{Fwdch(t)} = \begin{cases} \int_t^{t+\Delta t} ((P_{PV(t)} + P_{WT(t)}) - P_{L(t)}) dt & \text{if } (P_{PV(t)} + P_{WT(t)}) - P_{L(t)} > 0 \\ 0 & \text{if } (P_{PV(t)} + P_{WT(t)}) - P_{L(t)} \leq 0 \end{cases} \quad (3)$$

$$P_{Bdch(t)} + P_{Fwdch(t)} = \begin{cases} \int_t^{t+\Delta t} (P_{L(t)} - (P_{PV(t)} + P_{WT(t)})) dt & \text{if } (P_{PV(t)} + P_{WT(t)}) - P_{L(t)} < 0 \\ 0 & \text{if } (P_{PV(t)} + P_{WT(t)}) - P_{L(t)} \geq 0 \end{cases} \quad (4)$$

The proposed HRES in this research paper includes photovoltaic modules, wind turbines on the supply side, and hybrid energy storage made up of a Li-ion battery and flywheel. Using hybrid storage systems in this study allows efficient storage complementation, increased duty cycles, and higher storage capacity. The configuration of the proposed hybrid energy storage system (HESS) is shown in Fig. 2. A common DC-link bus is used to reduce the cost and associated power losses [22]. Here the battery and flywheel energy storage systems are connected to the same bus (DC-link), eliminating the requirement for a DC-AC inverter for flywheel energy storage system (FESS), subsequently reducing the cost and associated power losses significantly [22]. The research illustrates the flywheel's effective spinning reserve for efficient backup power to the load eliminating the use of diesel generators (DGs), which are prone to carbon dioxide (CO₂) emissions. In addition, the flywheel adequately helps increase the battery's lifetime by augmentation of more capacity and reducing battery degradation. The use of FESS further serves to reduce possible adverse frequency swings in the system [23].

HRES size optimization modeling:

Solar and wind resources have inadequate power production proficiency due to their intermittency nature and high dependence on natural weather. On the other hand, the correct system size requires an accurate balance between power generation and demand. This means precision is vital in designing the solar/wind/battery/flywheel system to meet the electrical load demand while optimizing the cost and reliability of power. This study, therefore, develops a multi-objective optimization approach to determine the system's size, cost, and optimum reliability. The optimum solution enables desired decision in ensuring higher power reliability at affordable costs.

Multi-objective optimization:

The formulation of optimal system size is based on the cost of all

components, installation, and operational costs. The optimum power reliability is based on the possible minimum cost of electricity. During optimization, two objectives with conflicting goals display the best compromise for the system's COE and reliability index.

The mathematical formulation for the multi-objective optimization model, which brings out a combined cost and reliability objective functions, is expressed as follows:

$$\text{Min}F(x) = \{f_1(\text{COE}), f_2(\text{UME})\} \quad (1)$$

This is subject to the following technical constraints:
Supply-demand balance constraint:

$$P_{PV(t)} + P_{WT(t)} + P_{Bdch(t)} + P_{Fwdch(t)} = P_{L(t)} + P_{Bch(t)} + P_{Fwch(t)} \quad (2)$$

Battery and Flywheel charging and discharging constraints:

The total capital cost and annual operation and maintenance costs are summed, putting into consideration the Levelized Cost of Electricity generation (LCOE) from the proposed HRES as follows:

$$\text{COE} = \left(\frac{IC_i + RC_i + OMC_i}{(1+r)^y} \right) / \left(\frac{P_{Gen}}{(1+r)^y} \right) \quad (5)$$

The annual power generation, P_{Gen} from the HRES is expressed in the equation:

$$P_{Gen} = \sum_{t=1}^{8760} (P_{PV(t)} + P_{WT(t)} + P_{Bdch(t)} + P_{Fwdch(t)}) \quad (6)$$

The total annualized cost of the HRES can be calculated as follows:

$$IC = \sum_{t=1}^{8760} (P_{PV(t)} IC_{PV} + P_{WT(t)} IC_{WT} + P_{FW(t)} IC_{FW} + P_{bat(t)} IC_{bat}) \quad (7)$$

The unmet electricity index is considered in this research to measure the reliability aspect of the HRES system, which can be defined as follows:

$$\text{UME}_{(t)} = \sum_{t=1}^{8760} \left\{ \frac{(P_{Bdch(t)} + P_{Fwdch(t)} + P_{L(t)} - P_{WT(t)} - P_{PV(t)} - P_{Bch(t)} - P_{Fwch(t)})}{P_{Gen(t)}} \right\} \quad (8)$$

The ability to minimize possible power inadequacies is virtuously anchored on the dynamic power-balancing capability of the system while ensuring reasonable economic implications (Camargo et al., 2019). To measure the reliability aspect of the HRES system, the Energy Index of Self-Reliance EISR is introduced based on the Markov model for reliability for a two-unit system [24]. In this case, the UME is a time-dependent ratio representing the total unmet electricity demand for the hybrid system, considering two-unit states of power generation for PV and wind turbine and the amount of the discharged electricity to satisfy a residential electrical load demand. The Energy Index of Self-

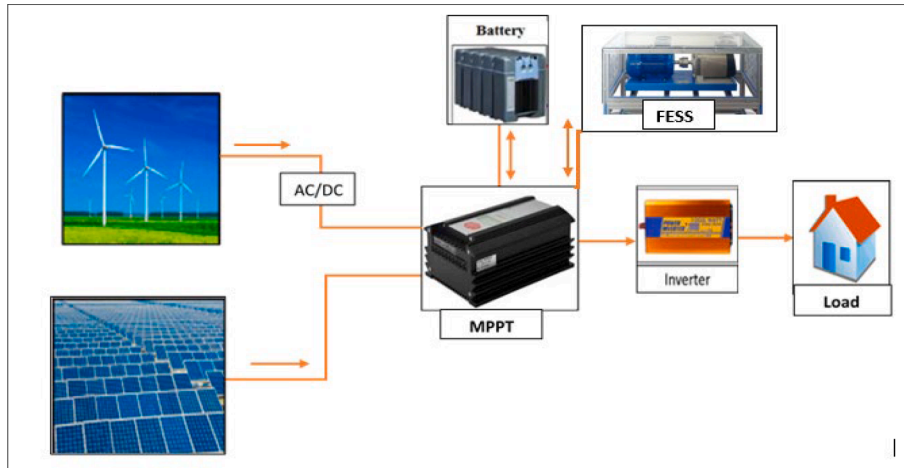


Fig. 2. Schematic diagram of the proposed HRES in this study.

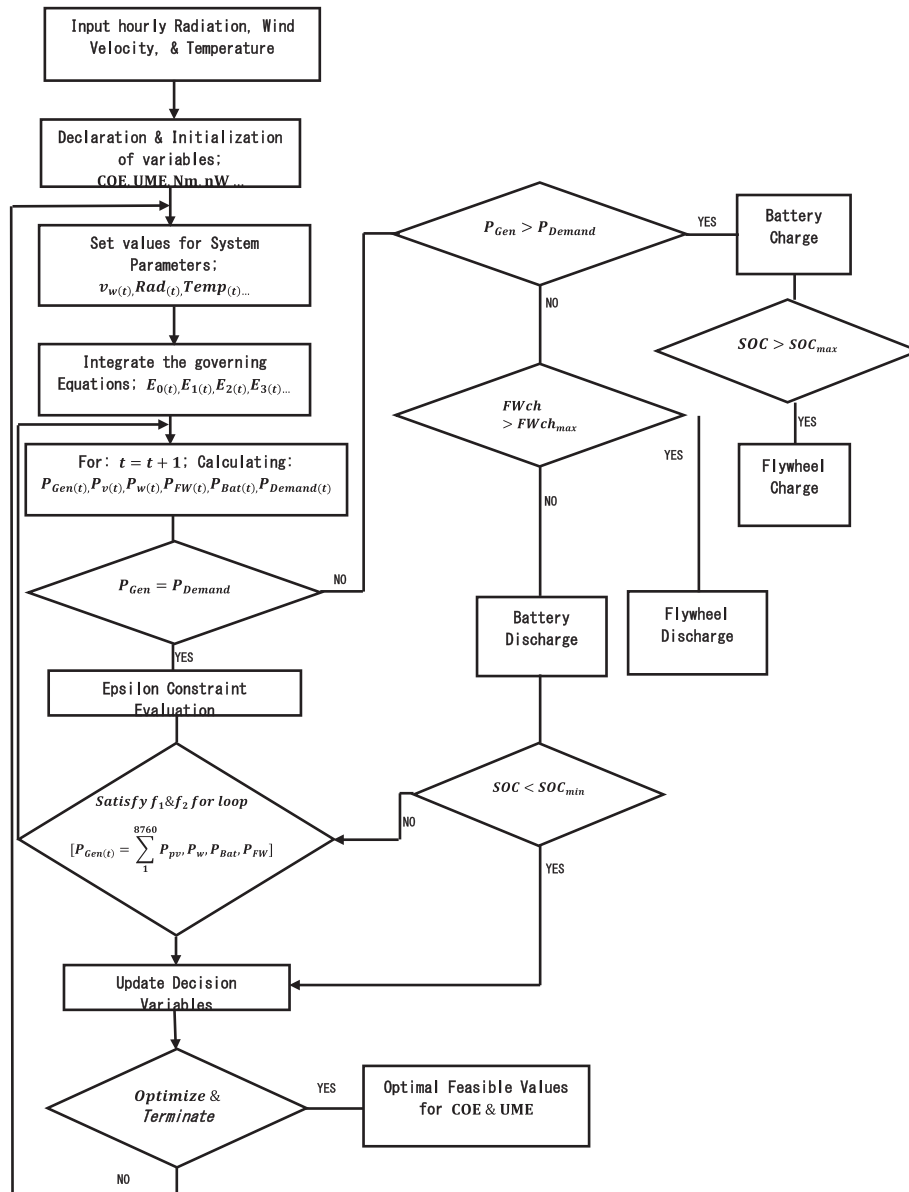


Fig. 3. Optimization flow chart.

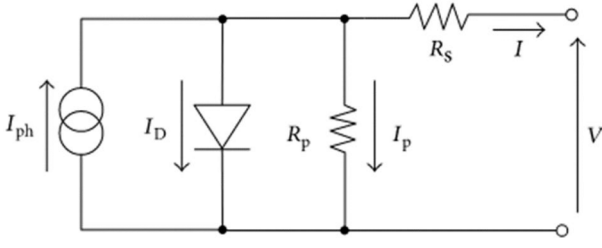


Fig. 4. Equivalent circuit of a PV module.

Reliance EISR is then introduced, using the following equation:

$$IESR_{(t)} = 1 - UME_{(t)} \quad (9)$$

The value of EISR is always less than unity; when EISR = 1, this signifies the ideal or perfect reliability of the HRES system.

The ϵ -Constraint method is adopted as an efficient and modest optimization strategy for solving the multi-objective challenge. In this technique, one of the objective functions is selected, and the second objective function transforms to act as a restriction considering a specific constraint, epsilon (ϵ) [25]. Introducing different values of ϵ , results for optimal Pareto solutions are obtained through a mixed-integer algorithm in a non-linear environment, generating Pareto feasible and non-dominated solutions [26].

The functions $f_{1(COE)}$ and $f_{2(UME)}$ are treated as constraints in the epsilon-constraint method in sequence. Hence, the model optimizes one objective function while keeping the other objective function within certain set proximities of deviation. For an allowable selected range of ϵ ; Therefore, the multi-objective optimization model can be re-written in the following format:

$$Min f_{1(COE)} \quad (10)$$

Subject to:

$$\begin{cases} f_{2(UME)} \leq \epsilon_1 \\ \vdots \\ f_{kmin} \leq \epsilon \leq f_{kmax} \end{cases} < \text{Technical constraints} >$$

where, f_{kmin} and f_{kmax} represent a range of all objective functions formulated due to adopting changing values of ϵ and therefore used as constraints. This creates a lexicographic series of objective function optimization where the initial objective function solution is obtained among the possible alternative optimal solutions, and the algorithm optimizes for the second or subsequent objective function [27]. Thus, in every optimum solution where $f_{1(COE)} = x_1$ for a selected epsilon value ϵ_1 , there is a corresponding $f_{2(UME)} = x_1^*$ for values in the range of $f_{kmin} \leq \epsilon_1 \leq f_{kmax}$.

The multi-objective optimization algorithm in the General algebraic modeling system (GAMS) is demonstrated in the flowchart in Fig. 3.

Mathematical modeling of the HRES components:

Solar Photovoltaic:

The intensity of solar radiation and ambient temperatures are important factors in PV power generation. The power generated by the PV module in this research is modeled using the equation below:

$$P_{pv(t)} = N_m V_{max(t)} I_{max(t)} \quad (11)$$

The number of PV modules N_m , is determined during the optimization process while $V_{max(t)}$ and $I_{max(t)}$ are the values of the maximum voltage and current of the module at time t , obtained using the PV circuit

equations as follows:

$$I = I_{sc} \left[1 - \exp \left(\frac{V - V_{oc} + IR_s}{V_{th}} \right) \right] \quad (12)$$

The current I is the equivalent circuit current for the module shown in Fig. 4. Equation (12) illustrates the relationship between the short circuit current, I_{sc} , open circuit voltage, V_{oc} , series resistance R_s , and thermal voltage V_{th} . Series resistance is calculated using the expression below [28].

$$R_s = \left[\frac{V_{oc} - V_{m(STC)} + V_{th} \ln \left(1 - \frac{I_{m(STC)}}{I_{sc(STC)}} \right)}{I_{m(STC)}} \right] \quad (13)$$

The values for the short circuit current I_{sc} and open circuit voltage V_{oc} are dependent on the irradiance G and obtained from the expressions below:

$$I_{sc} = I_{sc(STC)} \frac{G}{G_{(STC)}} [1 + \mu_{I_{sc}} (T_c - T_{c(STC)})] \quad (14)$$

$$V_{oc} = V_{oc(STC)} + \mu_{V_{oc}} (T_c - T_{c(STC)}) \left(1 + \sigma_{oc} \ln \left(\frac{G}{G_{(STC)}} \right) \right) \quad (15)$$

Wind turbine:

The power generated by the wind turbine is on accounts for the conversion of the wind's kinetic energy to electrical energy [29]. This happens in three stages; the aerodynamic stage, in which moving air is converted into mechanical energy, and the mechanical stage comprising torque speed transforming into an electrical power constituent. The functional factors here are turbine size, wind speed, and the height of the rotor [30]; the relationship is explained using the following equations:

$$v = v_{ref} \left(\frac{h_{hub}}{h_{ref}} \right)^\alpha \quad (16)$$

The wind speed v is directly proportional to the wind speed at the reference height, v_{ref} and the turbine height h_{hub} , while the reference height h_{ref} is inversely proportional. The power law coefficient α , fluctuates between 0.1 to a value equal to or slightly higher than 0.25 in cases of flat land to highly vegetative areas duly determined by terrain, height, season, wind speed variations, day, and hour within a specified geographical context. The power generated by the wind turbine at the time, t ($P_{WT(t)}$) is obtained as below [31]:

$$P_{WT(t)} = \begin{cases} 0 & v_{cut.out} < v_{(t)} < v_{cut.in} \\ v_{(t)}^3 \left(\frac{P_{rated}}{v_{rated}^3 - v_{cut.in}^3} \right) - P_{rated} \left(\frac{v_{cut.in}^3}{v_{rated}^3 - v_{cut.in}^3} \right) & v_{cut.in} \leq v_{(t)} < v_{rated} \\ P_{rated} & v_{rated} \leq v_{(t)} < v_{cut.out} \end{cases} \quad (17)$$

The rated power of the turbine is achieved at v_{rated} , this is a suitable compromise between $v_{cut.in}$ at which generation starts and $v_{cut.out}$ when the turbine shuts down to avoid damage [30].

Energy storage system:

Renewable energy microgrids are more reliable when coupled with a sound and efficient energy storage system. This is due to variations and unpredictability in solar radiation and wind speeds. When the energy produced by the HRES, $E_{gen(t)}$ surpasses the electrical load, $E_{L(t)}$, the surplus energy is stored in battery bank storage according to equation (18) stated below [17].

$$E_{bat(t)} = E_{bat(t-1)} + (E_{gen(t)} - E_{L(t)}) \quad (18)$$

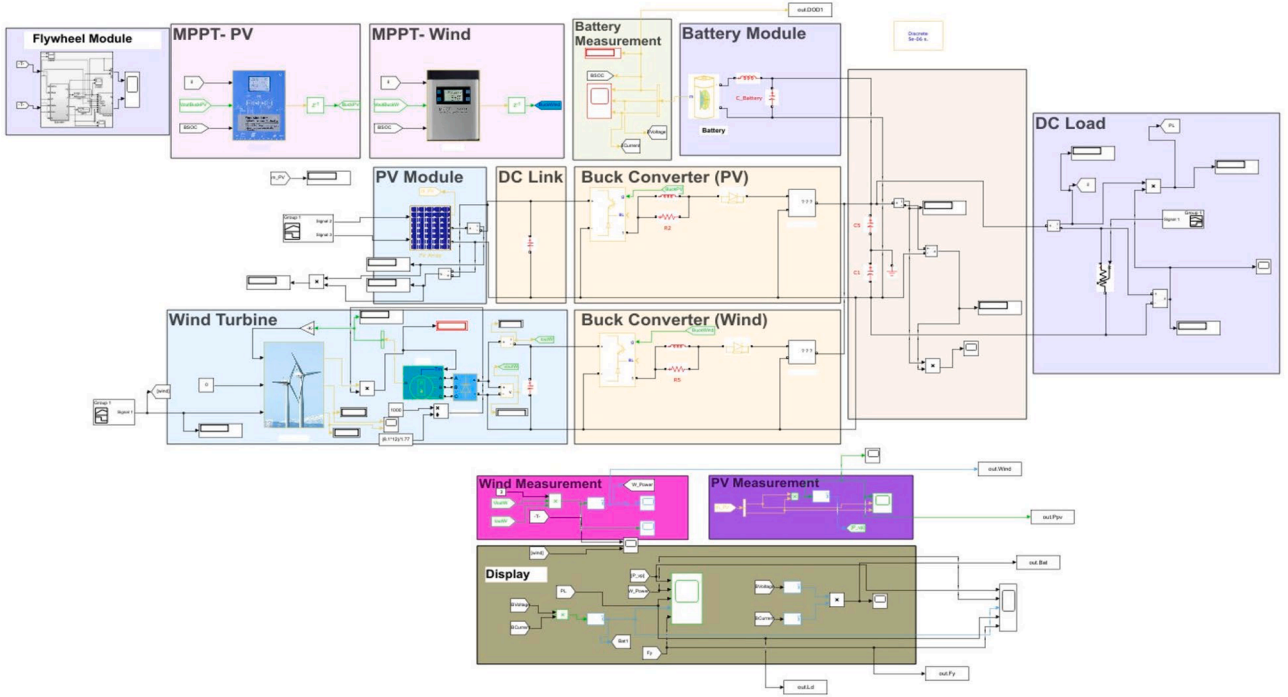


Fig. 5. Schematic diagram of HRES System in Matlab/Simulink.

where $E_{bat(t)}$ is the energy stored in the battery at time t , $E_{bat(t-1)}$ is energy stored in the battery at the time $(t-1)$, (Zhang et al., 2021). At any time, t the value of E_{bat} satisfies the constraint:

$$E_{bat_min} \leq E_{bat(t)} \leq E_{bat_max} \quad (19)$$

Using the charge transfer in and out of the battery, the SOC will be examined (Zhang et al., 2021; [32]).

$$SOC(t) = SOC(t_0) - \frac{1}{Q} \int_{t_0}^t I_{bat(t)} dt \quad (20)$$

$$I_{bat} = \frac{P_{Bdch}}{V_{bat}} \quad (21)$$

Flywheel is a kinetic-dependent energy storage technology [33]. Energy is stored in a rotating mass depending on the speed of the rotor mass, shape, and angular speed of revolution, which is expressed in equation (22).

$$E_{Fw} = \frac{1}{2} I w_f^2 \quad (22)$$

Here, E_{Fw} is the stored kinetic energy, I is the moment of inertia, and w_f is the angular velocity of the flywheel. The moment of inertia is a function of the mass of rotor and the rotor shape factor, therefore; outer

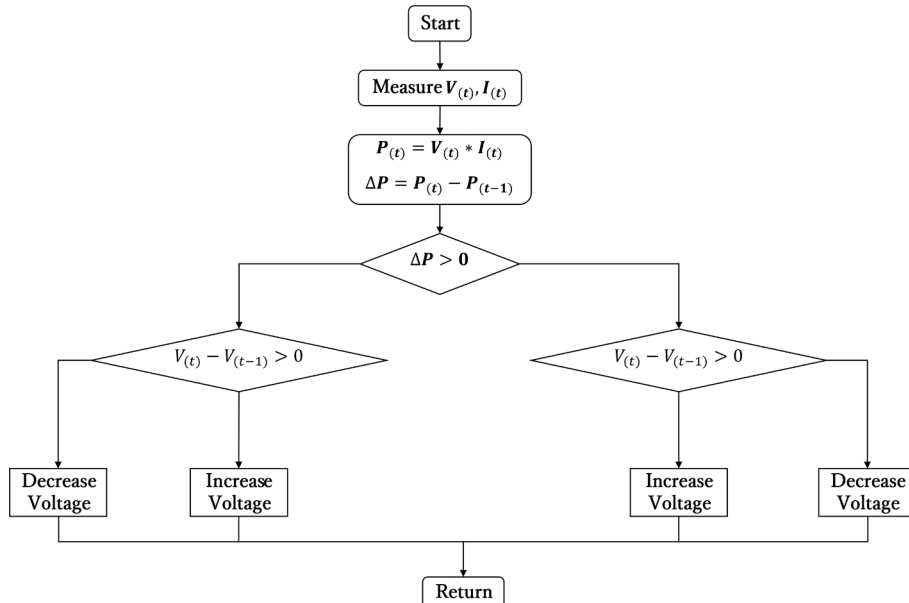


Fig. 6. P & O MPPT algorithm flowchart.

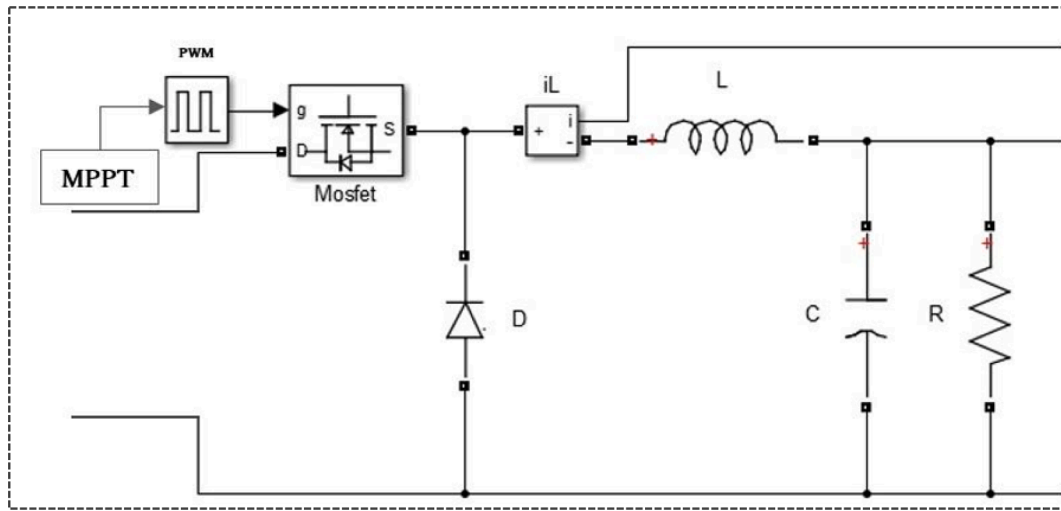


Fig. 7. DC-DC buck converter control circuit.

radius r_o , inner radius r_i , height h and the mass density ρ relate according to equation (23).

$$E_{Fw} = \frac{1}{2} \pi \rho h \omega_f^2 (r_o^2 - r_i^2) \quad (23)$$

The flywheel safely operates within ω_{fmax}^2 and ω_{fmin}^2 to limit voltage variation and maximum torque for a given power rating [34]. Therefore, the useful energy of a flywheel is defined within maximum and minimum angular velocity, ω_f :

$$P_{Fw(t)} = \frac{1}{2} I (\omega_{fmax}^2 - \omega_{fmin}^2) \quad (24)$$

HRES dynamic operation and control modeling:

The HRES Matlab simulation architectural structure in Fig. 5 utilizes a hybrid model of P&O MPPT to ensure optimal power output from the proposed HRES. The output power satisfies the demand via a DC-DC buck converter and further speeds up the charging of the HESS.

Perturb & observe algorithm for MPPT technique

The maximum power point is tracked using a minor perturbation that causes the power variation of the solar module. From time to time, the PV output is measured and compared with the previous power value. This is achieved by obtaining the instantaneous voltage $V(t)$ and current $I(t)$ for the calculation of power $P(t)$ at time t . This value is compared to

the preceding output value $P(t-1)$ measured at the time, $(t-1)$. The perturbation algorithm adjusts the operating MPP according to whether the positive or negative difference is obtained as described in Fig. 6. This process is continuously repeated, varying the DC/DC converter's duty cycle to obtain a maximum power point. A PI controller is coupled to a limiter lowering chances of over-saturation during the generation of carrier pulse for the IGBT switch.

DC-DC buck converter

The buck converter operates by step-down conversion of the applied dc input signal. The MOSFET or IGBT chopper circuit acts as a switch with a controlled switch, S. the inductor stores energy, ultimately delivering it to the load resistor. The power MOSFET or IGBT is turned off to maintain zero potential between the gate to the source terminal or to the collector terminal of the IGBT; the on-and-off variations result in a low-pass filter for the reduction in the output voltage [4]. The low-pass LC filter is perfect for the generation of regulated dc output [35].

The resistor, R, in the schematic DC-DC buck converter circuit in Fig. 7, provides a voltage drop to the circuit. The action process is twofold. Firstly, the switch is set to the ON condition, which facilitates the flow of current. The inductor, L and the capacitor, C, provide energy storage, subsequently allowing for the voltage drop across the load. The energy storage by the inductor leads to the application of Lenz's law by opposing the cause of its production, resulting in a reversal of polarity across the inductor due to induced current [4]. The total time, T

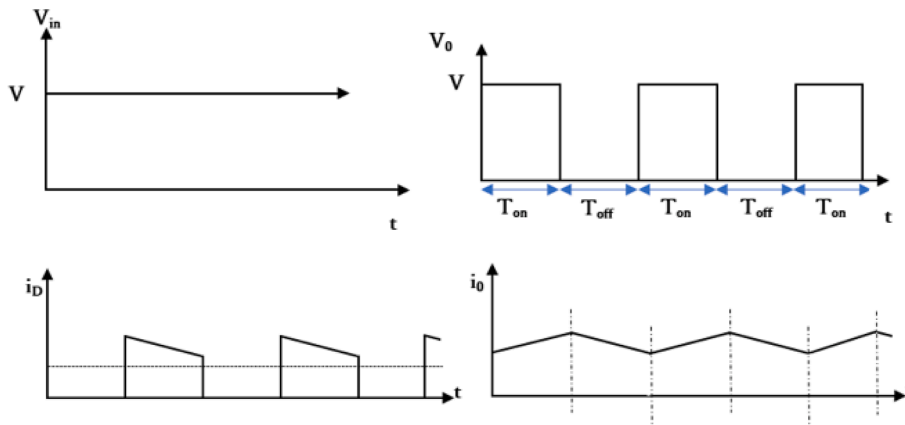


Fig. 8. Buck converter operation waveforms [35].

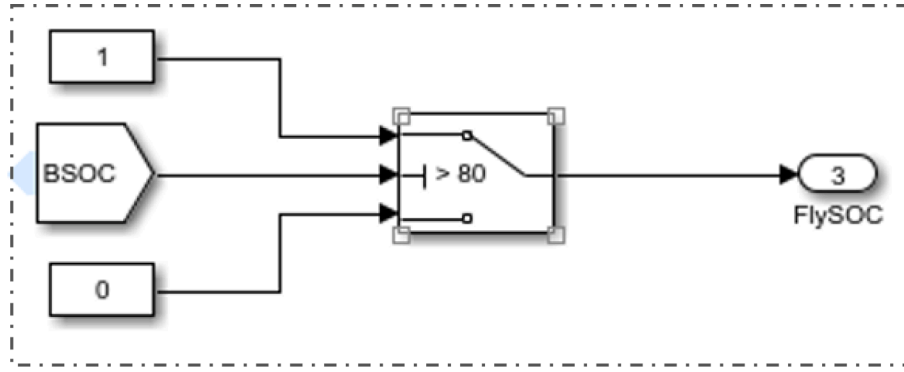


Fig. 9. Flywheel-battery charge control scheme.

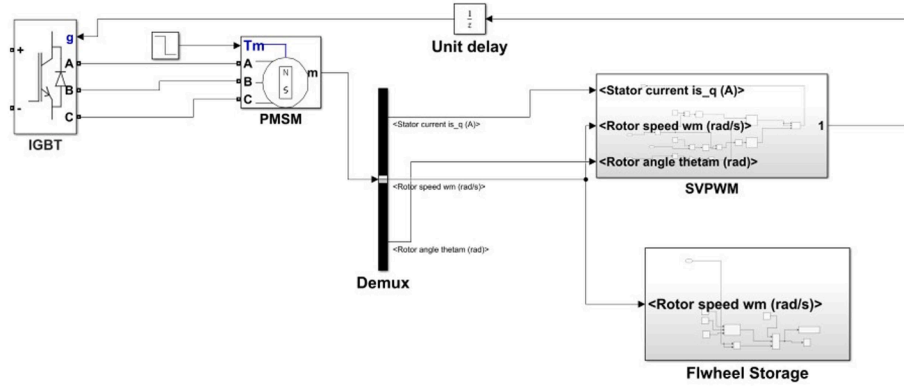


Fig. 10. Matlab/Simulink flywheel model.

combination is expressed in terms of T_{on} and T_{off} such that $T = T_{on} + T_{off}$ and the duty cycle is given by, $D = \frac{T_{on}}{T}$. When the switch is closed, therefore the expression for the change in current becomes:

$$\Delta i_t = \left(\frac{V_{in} - V_0}{L} \right) T_{on} \quad (25)$$

The second stage takes place when the switch is set in the OFF state. Again, the inductor acts as a source because it has stored energy. However, owing to reversed polarity by the inductor of the diode, D is in the forward-biased state. Therefore, the current is allowed to flow in the opposite direction until the inductor stored energy gets depleted. This makes the diode reverse-biased again, allowing the switch to be closed to start the cycle again automatically. The current change rate through the inductor during the second mode of operation can be represented using the expression.

$$\Delta i_L = \left(\frac{V_0}{L} \right) (1 - D) T \quad (26)$$

The two modes represent a steady state operation with a recurring waveform, shown in Fig. 8 [35].

$$\int_0^T V_L dt = \int_0^{T_{on}} V_L dt + \int_{T_{on}}^T V_L dt = 0 \quad (27)$$

Battery-Flywheel charge control method:

Flywheel storage systems have a high energy density, which allows them to be cycled frequently with less impact on performance. They are, therefore, suitable for applications requiring frequent cycling since they incur limited life reduction on excessive use. On the other hand, battery storage supports high energy loads but at low power. Therefore, our

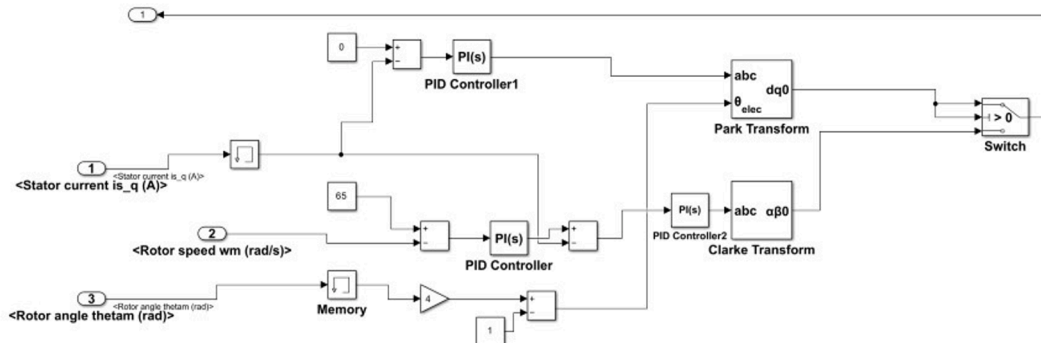


Fig. 11. SVPWM subsystem in flywheel model.

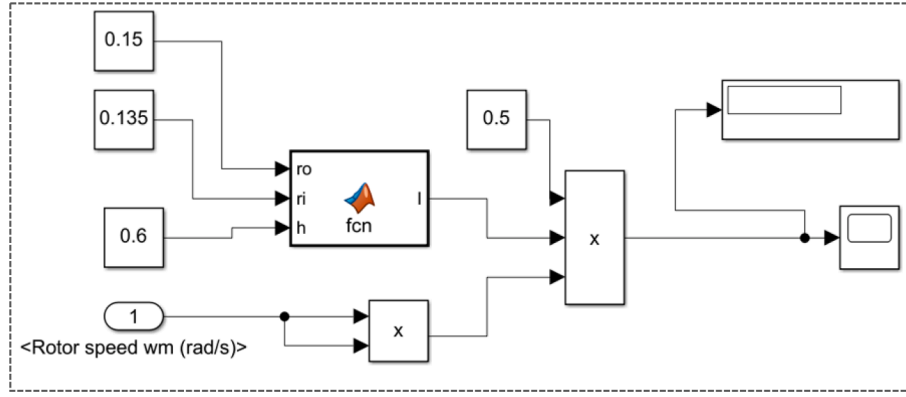


Fig. 12. Flywheel energy storage subsystem.

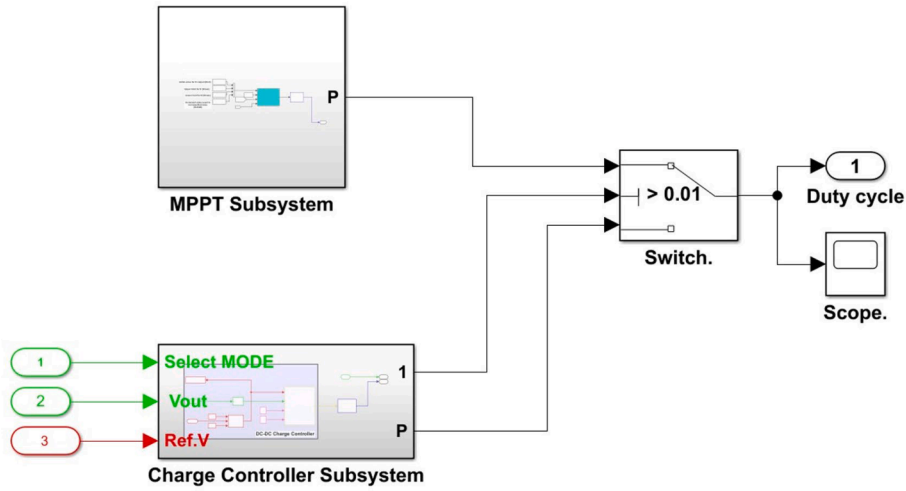


Fig. 13. Battery charging model [35].

charging control methodology prioritizes battery charging and flywheel discharge. Fig. 9 shows the control strategy for flywheel charging only when the battery state of charge (SOC) is above 80 %. This ensures charging priority is given to the battery in the event of excess power output from the PV-Wind power generators.

The flywheel uses the electromechanical principle to store energy [23]. A motor is used to convert electrical energy from the source into mechanical energy. The same motor is used as a generator during flywheel discharge to provide electrical energy. The flywheel model in Matlab/Simulink is shown in Fig. 10. It is composed of a permanent magnet synchronous machine, space vector pulse width modulation, and an energy storage subsystem.

The motor/generator is modeled as a permanent magnet synchronous machine (PMSM). The electrical system is expressed using the equations:

$$\frac{d\psi_{sd}}{dt} = -V_{sd} - R_s I_{sd} - \omega_m \psi_{sq} \quad (28)$$

$$\frac{d\psi_{sq}}{dt} = -V_{sq} - R_s I_{sq} - \omega_m \psi_{sd} \quad (29)$$

$$T_e = 0.5\rho(\psi_m I_{sq} + (L_d - L_q) I_{sd} I_{sq}) \quad (30)$$

The mechanical system is expressed by equations:

$$\frac{d\omega_m}{dt} = \frac{1}{J} (T_e - T_f - F_{\omega_m} - T_m) \quad (31)$$

$$\frac{d\theta}{dt} = \omega_m \quad (32)$$

Space vector pulse width modulation (SVPWM) applies a modulation scheme for a given voltage vector that emulates a three-phased sinusoidal waveform providing frequency and amplitude control. Using a Clarke-park transformation, we achieve a two-dimensional d-q frame. The “d” component represents the flux, and the “q” component represents the torque. PI controllers are used to obtain speed regulation and proportional plus integral PI for current regulation. The SVPWM subsystem is shown in Fig. 11. Energy storage in the flywheel is illustrated by equation (22), explained earlier. The flywheel energy storage subsystem is illustrated in Fig. 12.

Fig. 13 represents the dynamic battery charge and discharge process in Simulink. Equations (33) and (34) represent the battery mode selection.

Discharge model ($i^* > 0$):

$$E_{charge1}(it, i^*, i, Exp) = E_0 - K \cdot \frac{Q}{Q - it} \cdot i^* - K \cdot \frac{Q}{Q - it} \cdot it + Laplace^{-1} \left(\frac{Exp(s)}{Sel(s)} \cdot 0 \right) \quad (33)$$

Charge model ($i^* < 0$):

$$E_{charge2}(it, i^*, i, Exp) = E_0 - K \cdot \frac{Q}{it + 0.1 \cdot Q} \cdot i^* - K \cdot \frac{Q}{Q - it} \cdot it + Laplace^{-1} \left(\frac{Exp(s)}{Sel(s)} \cdot \frac{1}{s} \right) \quad (34)$$

The capacity of the battery does not change according to the

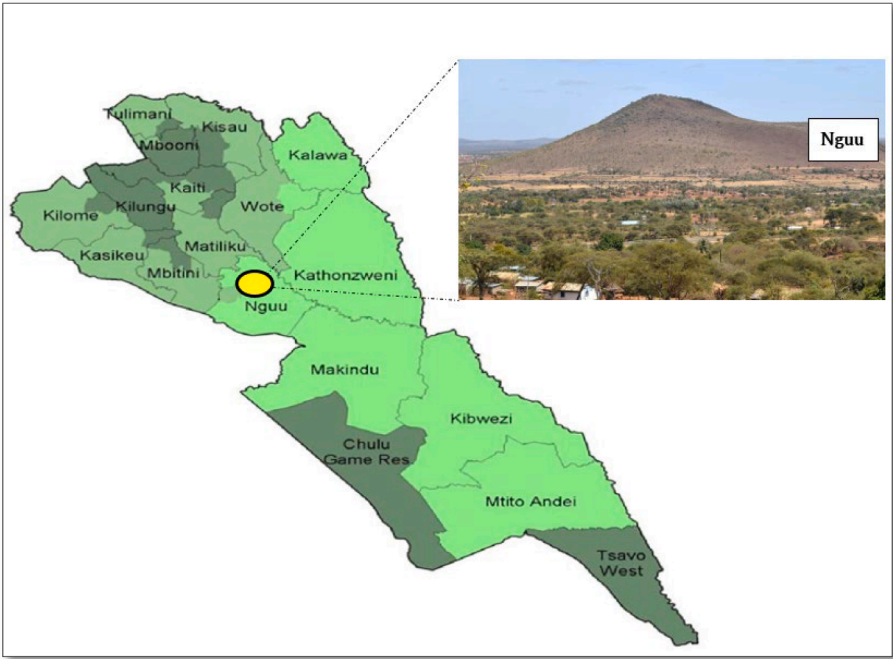


Fig. 14. Schematic focus on the local area of research (Makueni, Nguu).

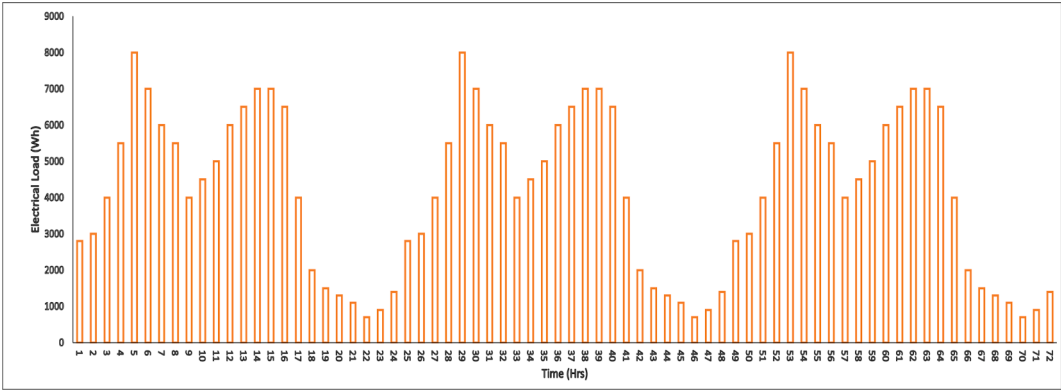


Fig. 15. Hourly Electrical load profile for 72 h in Makueni, Nguu.

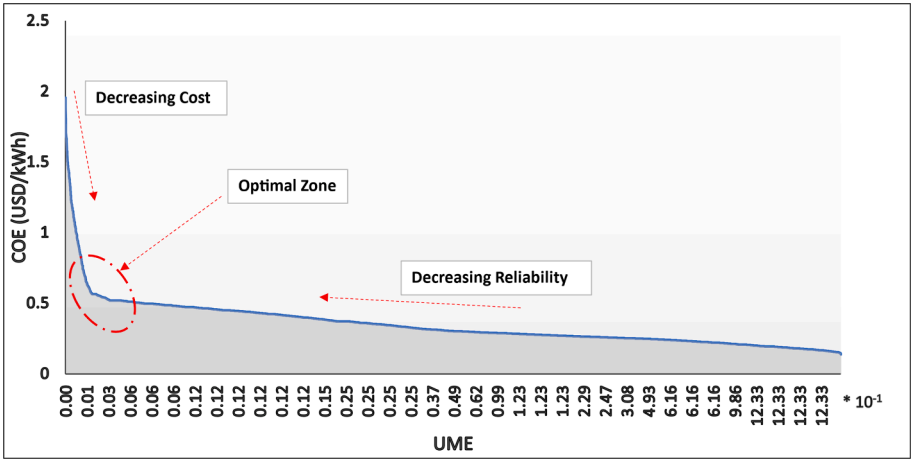


Fig. 16. Pareto frontier for the multi-objective ϵ -constraint solution.

Table 2
Optimization results.

Output Variable	COE	UME	Number of PV panels (@330Wp Rating)	Number of wind turbines (@1kW Rating)	Number of battery banks (@1kW Rating kWh)	Flywheel Capacity (kWh)
Optimal Value	0.519	0.003	26	3	22	4.7

Table 3
Selected optimization scenarios.

Scenario	Month	Period	Season
I	February	10th-12th day	warm/dry
II	April	23rd-25th day	long rain
III	June	7th-9th day	cold/dry
IV	September	14th-16th day	warm/dry
V	December	20th-22nd day	short rain

discharge current amplitude, according to the Peukert effect [4]. So that E_0 is constant voltage (V), $\text{Exp}(s)$ is the exponential zone dynamics (V), $\text{Sel}(s)$ is the battery mode (0 for discharge, 1 for charge), K is the

polarization constant (Ω), i^* is the low frequency current dynamics (A), i is the battery current (A), it is the extracted capacity (Ah), Q is the maximum battery capacity (Ah) and B are the the exponential capacity (Ah^{-1}). In this model, parameters are deduced from the discharging features and implied to the charging.

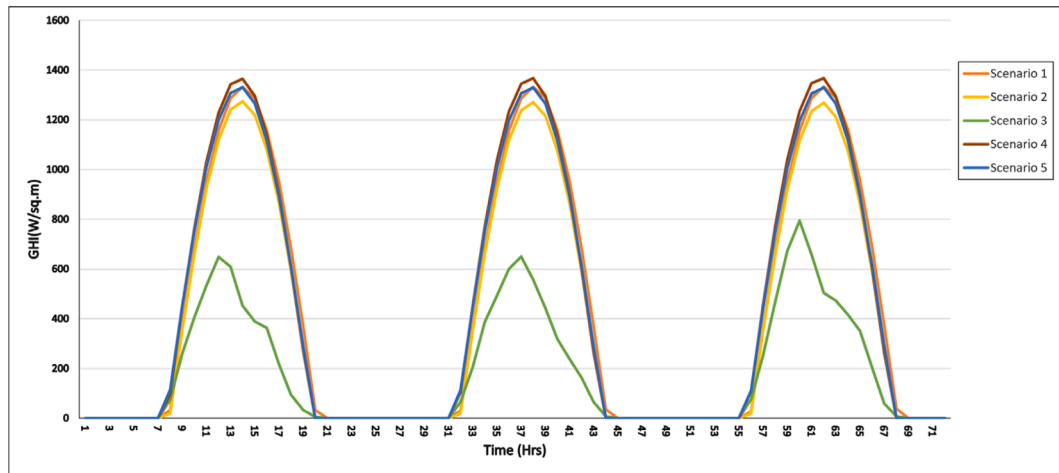
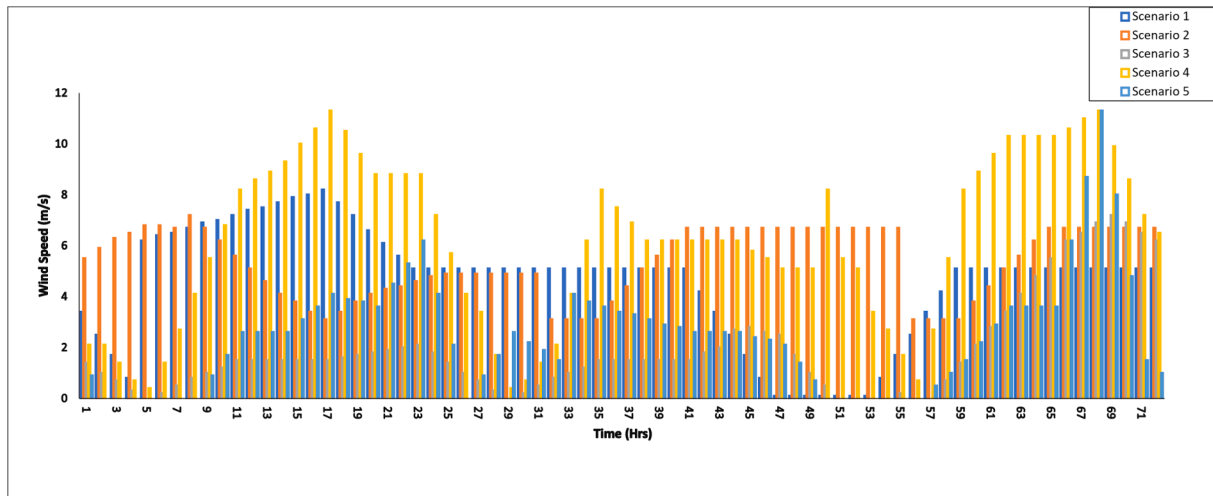
Results and discussion

Area of study

The annual data for the year 2021, solar radiation, ambient temperature, and wind speed, was collected from the Nguu area of Makueni county in Kenya, shown in Fig. 14 [36]. The electrical load profile was extracted from county government household energy consumption statistics and presented in Fig. 15 [37]. Using these data, different scenarios are optimized and verified to replicate an existing HRES power system.

Size optimization model results

In Fig. 16, an illustration of the multi-objective optimization Pareto front for the NLP ϵ -constraint solution is demonstrated. The values of COE and UME for the f_1 and f_2 objective functions are tabulated for each hour of the year (8760 h), giving a suitable compromise solution. This is because, a Pareto criterion exhibits a trading set-up in which neither side

**Fig. 17.** Radiation for the different scenarios.**Fig. 18.** Wind speed for different scenarios.

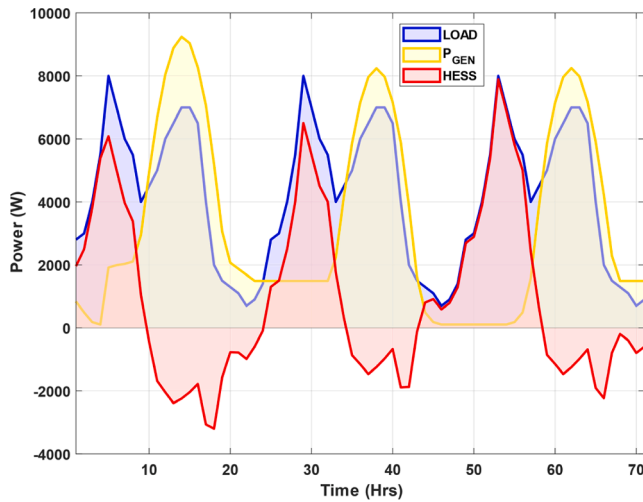


Fig. 19. Energy supply-demand balance for Scenario I.

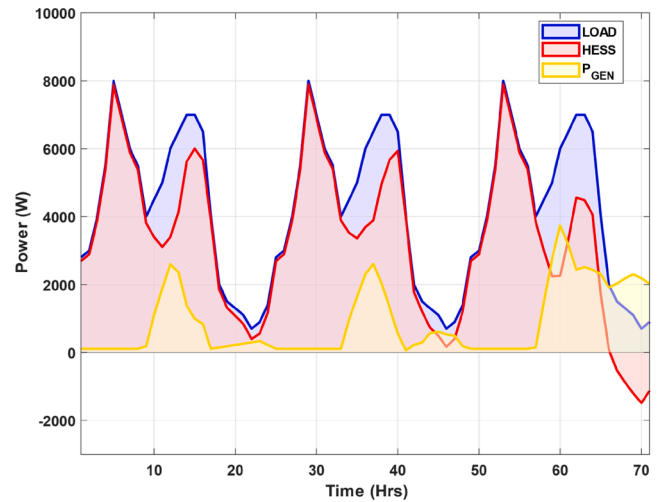


Fig. 21. Energy supply-demand balance for Scenario III.

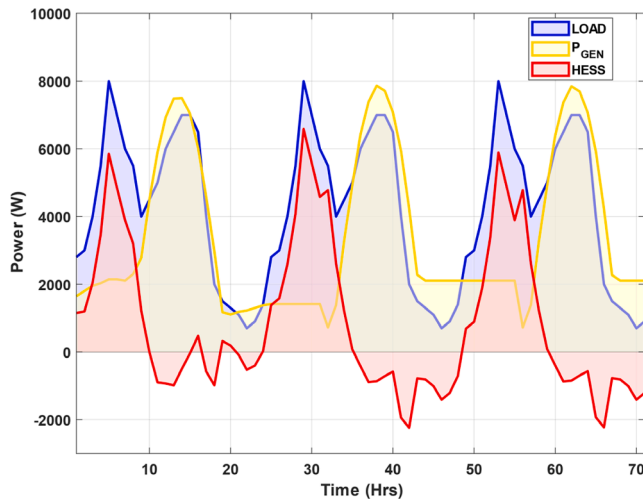


Fig. 20. Energy supply-demand balance for Scenario II.

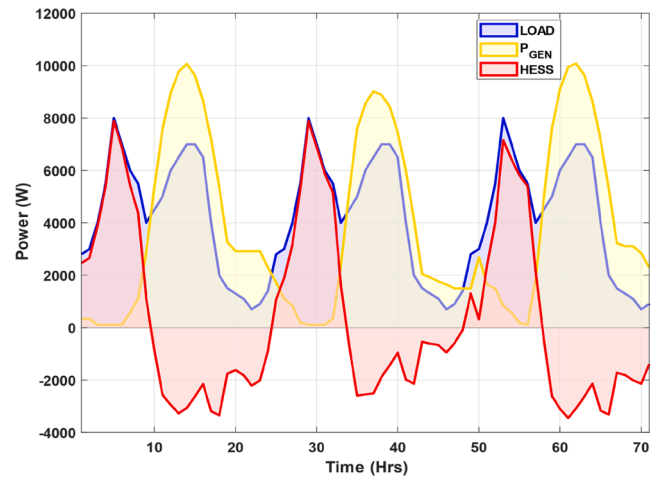


Fig. 22. Energy supply-demand balance for the Scenario IV.

of the function merits without inversely affecting the other function. Therefore, this provides a course-effect experience on impacts associated with any decision; for instance, a reduction in energy cost leads to possible energy unreliability, and the reverse is true, as described in Fig. 16. The iterations are looped for every hour of the year to produce a succession of optimum values for f_1 and f_2 appropriate in ensuring minimum cost of energy while avoiding possible blackouts in the HRES system. The value of the energy index of self-reliance, $EISR$ is 0.9969 when the UME value is 0.00308, representing a strong reliability index according to equation (9). While the values of COE decrease, the values of $EISR$ decrease, implying a reduced reliability state in the power supply if a lower cost of electricity is adopted.

The optimal results are summarized in Table 2. These results show a greater dependence on the use of PV as the most reliable power generation than wind turbines due to a high level of radiation in the makueni area and the low capital and installation costs of solar PV systems. In addition, battery storage is given more preference over flywheel; this also can be due to the high acquisition capital incurred in installing a flywheel storage system.

Scenario analysis

The five scenarios reported in Table 3 were considered to effectively represent the technical performance of the proposed HRES in order to deliver reliable energy throughout different seasons in Kenya.

The meteorological data used in analyzing the aforementioned scenarios are depicted in Fig. 17 and Fig. 18. The electrical load profile shown in Fig. 15 is used in all scenarios.

Scenario I:

In this scenario, the case study area experienced a clear sky with long sunny hours recording high irradiance of up to 1330 W/m^2 peak values at 1300–1400 hrs on all days. The wind profile shows high speeds recorded throughout day one from 9hrs to 18hrs with a peak value of 8.2 m/s . Fig. 19 shows that, the electricity generation decreases from 850 W at 0hrs to 109 W at 4hrs, then gradually increasing to 9.2 kW at 14hrs. The power output declines until 185 W at 45hrs. The HESS discharges from 0hrs to 10hrs to meet the load demand. At 10hrs, excess power is generated to charge the HESS. Generation is highest during the day and lowest at night when little or no radiation is received. The wind speed is lowest from 47hrs to 57hrs at 0.1 m/s to 2.5 m/s on the third day. The HESS discharge profile shows at 53hrs, a peak discharge of 8 kW is attained to satisfy the peak demand on the third day of the scenario. Unpredictable variations in wind speed led to heavy reliance on battery and flywheel storage systems.

Scenario II:

Scenario II represents the long rain season extracted from the

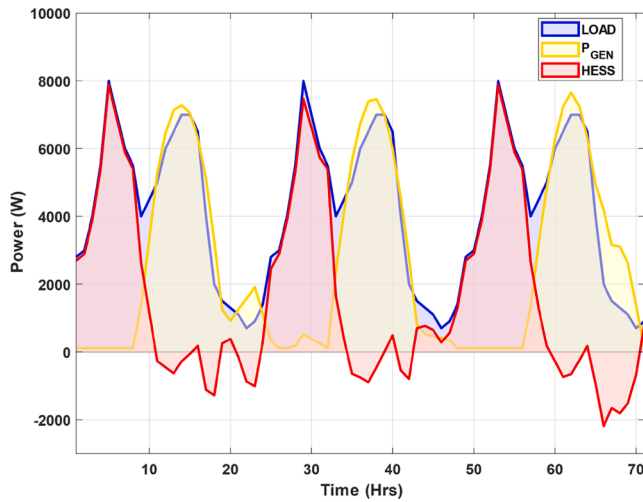


Fig. 23. Energy supply-demand balance for the Scenario V.

weather conditions from the 23rd to 25th days of April 2021. The maximum irradiance reaches 1270 W/m^2 between 13.00hrs–14.00hrs for all three days. The average wind speed is 5.3 m/s , and the maximum value is recorded at 7.2 m/s . The energy balance scenario shown in Fig. 20 depicts a similar power generation and HESS profile to scenario 1. However, in this scenario, since the load profile is similar, it can be observed that less power is generated to satisfy the load and charge the HESS. The maximum generation reaches 7.8 kW at 37hrs with a full charge capacity of 2.2 kW at 41hrs and 65hrs. During the day, solar radiation is high; hence, high PV power is harvested. However, the area experiences unpredictable wind speed during these three days leading to low generation output at night hours and, thus, high discharge levels at night.

Scenario III:

This period represents the lowest levels of irradiance and wind speed.

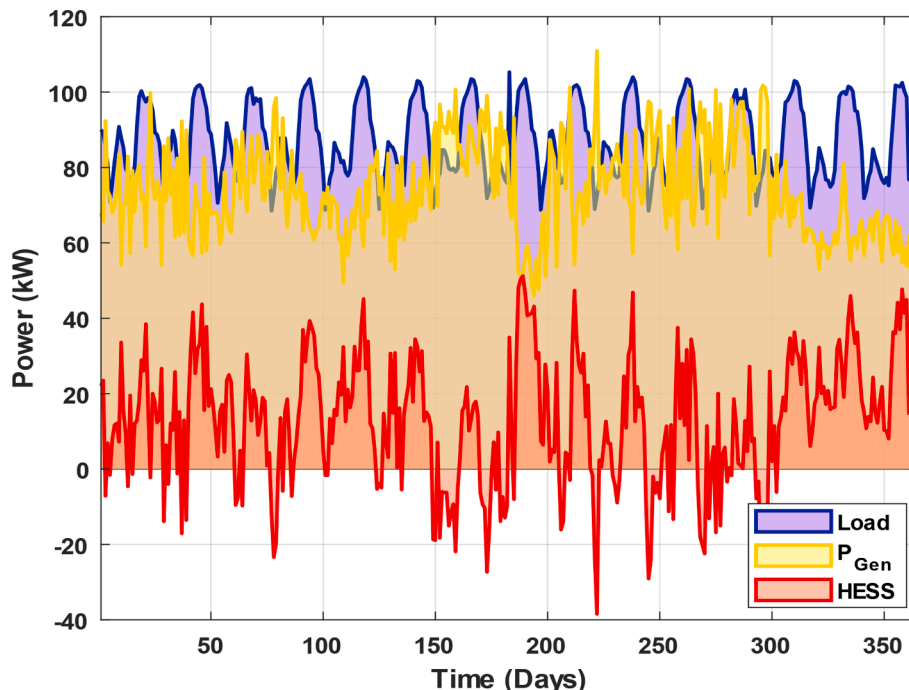


Fig. 24. Annual energy supply-demand balance for 2021.

The average wind speed is 1.9 m/s while the maximum irradiance is 790 W/m^2 at 60hrs on the third day. The results in Fig. 21 show that the HESS discharges from 0hrs to 66hrs when excess power is generated to charge the battery and flywheel storage.

Scenario IV:

Fig. 22 shows the energy balance obtained from diurnal optimization for meteorological conditions on the 14th to 16th days of September 2021. This month experiences the highest wind speed levels at solar irradiance, with a clear sky and high humidity with irradiance records of 1370 W/m^2 at midday. The maximum wind speed is 11.3 m/s with an average of 6.2 m/s . The HRES supply-demand balance curve in Fig. 22 shows during this scenario, the HESS discharges at 0hrs – 10hrs, 25hrs–34hrs, and 49hrs – 58hrs. These profiles represent night hours when the system depends entirely on wind speed for a generation. The graph shows 3.4 kW as the highest charge capacity attained at 61hrs on the third day. In this scenario, the maximum charge of the HESS is reached with less dependence on discharge due to favorable climatic conditions.

Scenario V:

In this scenario, the peak irradiance received during the midday hours of the 20th and 22nd days of December 2021 at 1330 W/m^2 . The wind speed average for this scenario is 2.4 m/s , with several hours of the three days recording 0 m/s wind speed levels. Fig. 23 represents the results of optimization for this scenario. According to the results, the system relies on HESS to satisfy the electrical demand at 0hrs to 10hrs, 25hrs to 34hrs, and 49hrs to 58hrs. This is accredited to low solar radiation during these hours and high negative variation in wind speed.

From these scenarios, it can be observed that the system is highly reliable, meeting the load demand sufficiently in all meteorological conditions experienced in the area of study. Fig. 24 shows an annual supply-demand balance curve for 2021, depicting 100 % satisfaction of the electrical load from PV-wind power generation and battery-flywheel discharge.

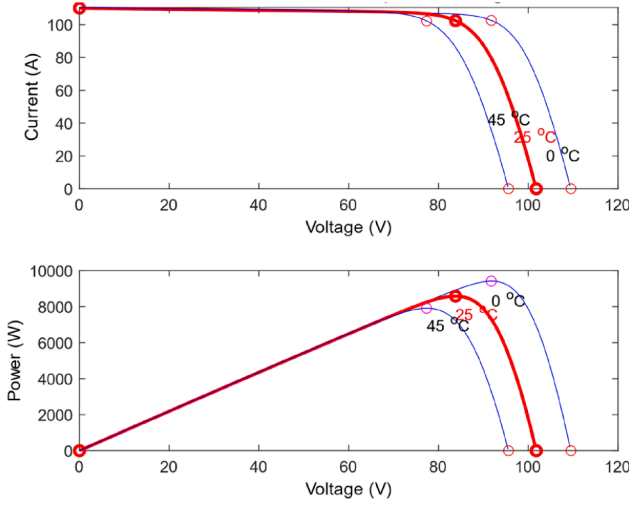
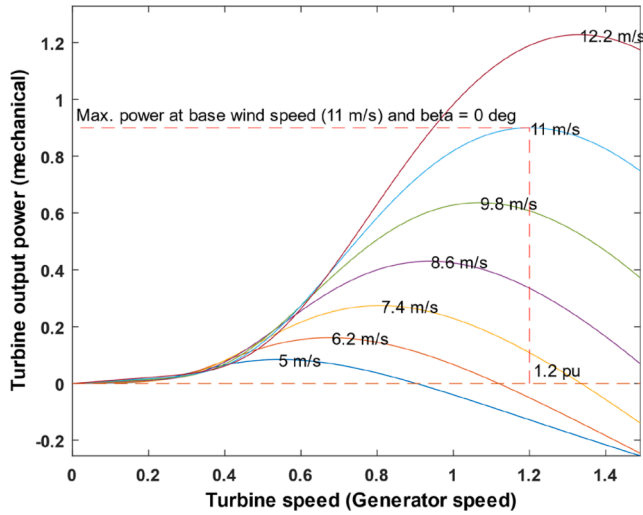


Fig. 25. I-V and P-V curves.

Fig. 26. Turbine Characteristics at $\beta = 0^\circ$

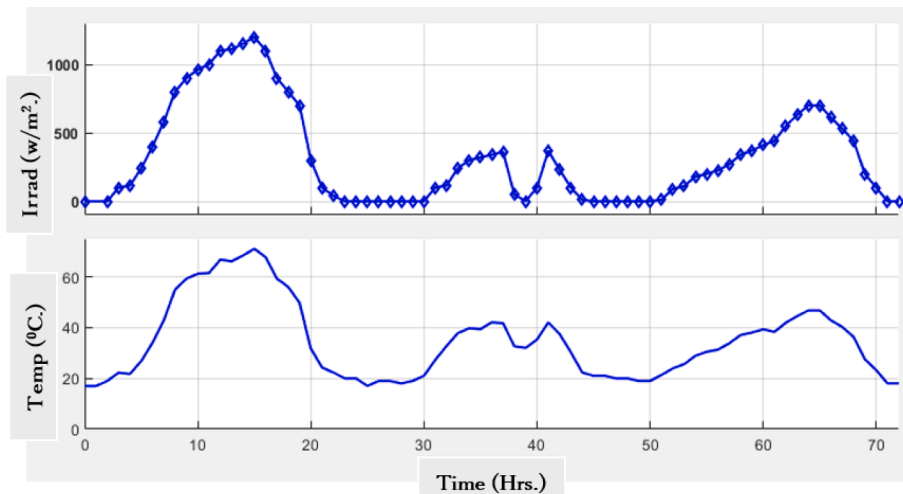
HRES dynamic power control results:

The estimated optimal size in the previous section was used in the Simulink model in order to evaluate the hourly performance of the proposed HRES under different operating conditions. The PV block has 13 parallel strings connected as two string pairs in series. The I-V and P-V characteristic curves for the PV system are demonstrated in Fig. 25 at 0°C , 25°C , and 45°C cell temperatures. To stabilize the DC voltage from the PV system, $500\mu\text{F}$ DC link capacitor is connected in parallel. The wind turbine is fed by wind speed, pitch angle, and generator speed. The turbine characteristics at $\beta = 0^\circ$ are shown in Fig. 26. Output torque per unit is fed to the PMSG to generate three-phase AC voltage, which is converted to DC voltage by the rectifier. A $500\mu\text{F}$ DC link capacitor is connected to the output. A lithium-ion battery type of 12 V/30Ah is used in this Simulink simulation. The initial state of charge is set at 60 %. The lithium-ion battery is fully charged at 14.4 V with a nominal discharge current of 2A and internal resistance of $0.014\ \Omega$. The parameter values for the proposed HRES components are detailed in Appendix.

Using Ohm's law, the power load demand for the community, as shown in Fig. 15, is converted to a dynamic resistive load in the Matlab-Simulink model for 72 h of operation. The wind speed, irradiance, and solar cell temperature data used in the simulation are represented in Fig. 27 and Fig. 28.

The simulation results in Fig. 29 illustrate the operation output for the HRES system for an arbitrary 72-hour scenario weather condition in Makueni during the month of June 2021. The MPPT charge controller is used to ensure the extraction of maximum power from PV and wind turbine performance by forcing maximum power point operation at fluctuating irradiance and wind speeds. The results show an effective power balance and perfect complementation between the electrical load demand and power generation and between charging and discharging operations based on the given optimal size of the HRES. The battery and flywheel charging are highlighted in this figure by a curve of negative gradient, while the discharge state is represented using a positive gradient curve.

An off-grid HRES has a major challenge in the supply and demand uncertainties that lead to low system inertia with possible variable frequencies. Such a state may expose the system to a complete blackout if measures are not implemented to balance supply with load demand. Diesel generators can play a critical role as a reserve, providing backup power to balance the load demand, though this option is considered detrimental due to the CO_2 emissions of diesel generators. However, in the proposed HRES, the flywheel energy storage system (FESS) alleviates possible blackouts due to frequency-based power imbalance. Based on the operation profile shown in Fig. 29, FESS is charged with positive

Fig. 27. Irradiance (W/m^2) and cell temperature ($^\circ\text{C}$).

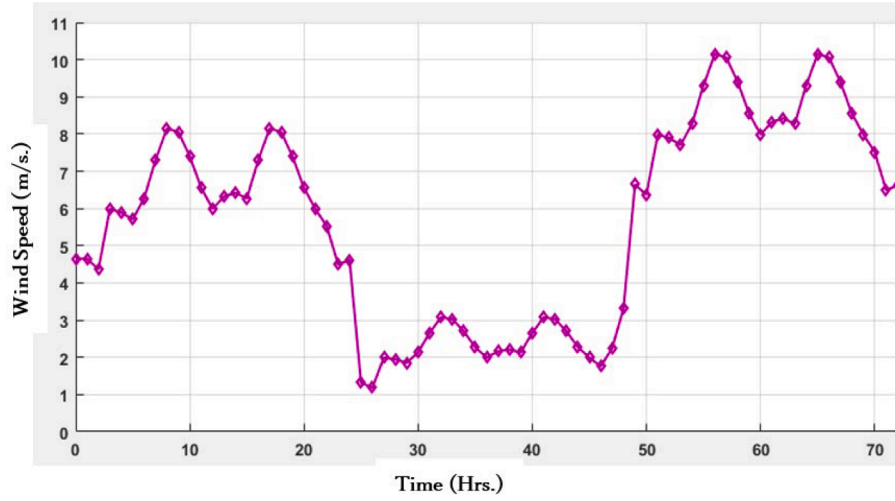


Fig. 28. Wind Speed in m/s.

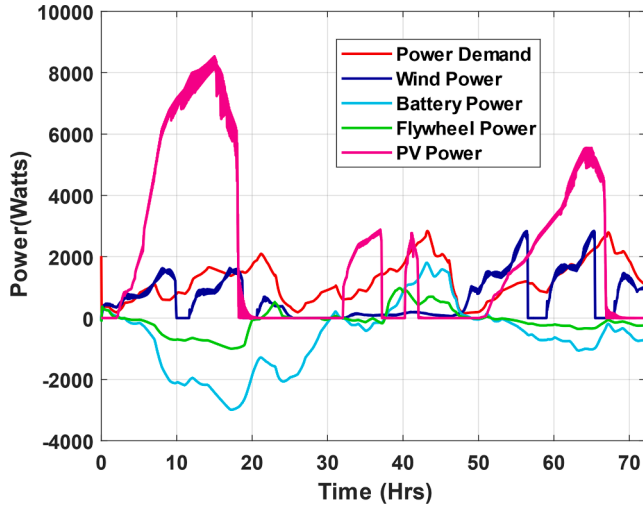


Fig. 29. Overall system performance for 72 h of operation of the proposed HRES.

torque generated upon excess power from the combined solar/wind output, wind power system, or PV power system. At any given moment, the flywheel charging lags the battery charge curve because the charge controller prioritizes battery charging. When the demand exceeds the total PV/Wind power output, the flywheel reverse-spins fast to respond to the abrupt deficit to satisfy the power deficiency in the HRES. This is demonstrated in magnified sections of Fig. 30. The battery response lags the flywheel, whose discharge curve, in comparison, tends to be positive promptly. This effectively substantiates the significance of the flywheel as an ideal spinning reserve for balancing the supply with electrical load demand.

Using a flywheel in the proposed HRES system further increases the battery storage capacity by a factor of 0.00886 % every 72 h of operation, as demonstrated in Fig. 31. The graph illustrates how the state of charge, (SOC) of the battery energy storage system (BESS) is affected by the flywheel operation in the system. The SOC is a major factor in determining the number of cycles of a battery within a given period of operation. This consequently underlines the aging characteristics of a battery system which ultimately dictates the lifespan and cost of storage systems. The visualization shows state of charge curve for the battery storage with and without a flywheel. In this case, the SOC of the battery with the flywheel maintains a higher charge percentage compared to

when the flywheel is not running. The most important observation here is how the loss of charge is minimal when hybridized storage (BESS/FESS) is used, contrary to SOC with stand-alone battery storage. The graph demonstrates a slowed discharge for hybrid Battery/Flywheel storage of 0.00886 % for 72 h of operation. This is equivalent to a 7045Wh cumulative storage capacity gain in 72 h of HRES operation and an 858kWh capacity increment for one year, as exemplified in Fig. 32. The use of the flywheel, therefore, provides an effective degradation mitigation system in which a substantial improvement in the life cycle and capacity of the battery is attained.

Calendar aging of the battery storage system versus capacity gain due to the hybridization of the battery and flywheel:

In this research paper, the hybridization of energy storage has been shown to effectively enhance the battery storage capacity of 0.00886 % for 72 h, equivalent to 7045Wh. However, it can be claimed that there is a degradation due to battery calendar aging by maintaining the SCO of the battery at a high level. The capacity loss of the battery is greatly influenced by temperature, time, state of charge (SOC), and charge/discharge rates [38]. The total capacity loss of the battery, Q_{loss} , is a function of both cycle and calendar losses, which can be expressed as below [39]:

$$Q_{loss} = Q_{loss}^{Cal} + Q_{loss}^{Cyc} \quad (35)$$

$$Q_{loss} = (a.T^2 + b.T + c) \exp[(d.T + e).I_{rate}] . Ah_{throughput} + f.t^{0.5} . \exp(-E_a/RT) \quad (36)$$

R is the gas constant, f and E_a represent the pre-exponential factor and activation energy which are approximated at 14,876 (1/day^{1/2}) and 24.5 (kJmol⁻¹) for lithium-ion batteries, respectively. The values of the charge/discharge rate coefficients (a, b, c, d , and e) are given by [38], [40]. These values are estimated experimentally to examine the aging and degradation of lithium-ion batteries due to temperature, depth of discharge (DOD), and charge/discharge rate. The operation time, t , is given in days. I_{rate} represents the charge/discharge rate (see Table A-1). The energy throughput, $Ah_{throughput}$, is given by the equation below:

$$Ah_{throughput} = Q(1 - SOC)n_{cyc} \quad (37)$$

n_{cyc} is considered to be one cycle per day. In order to simplify the calculation, the battery cell temperature is considered approximately the same as the ambient temperature.

Fig. 33 demonstrates the percentage capacity loss due to calendar and cycle aging for the HRES in 72 h of operation. The results show that the battery loses about 0.00094 % of capacity, equivalent to 715Wh. The

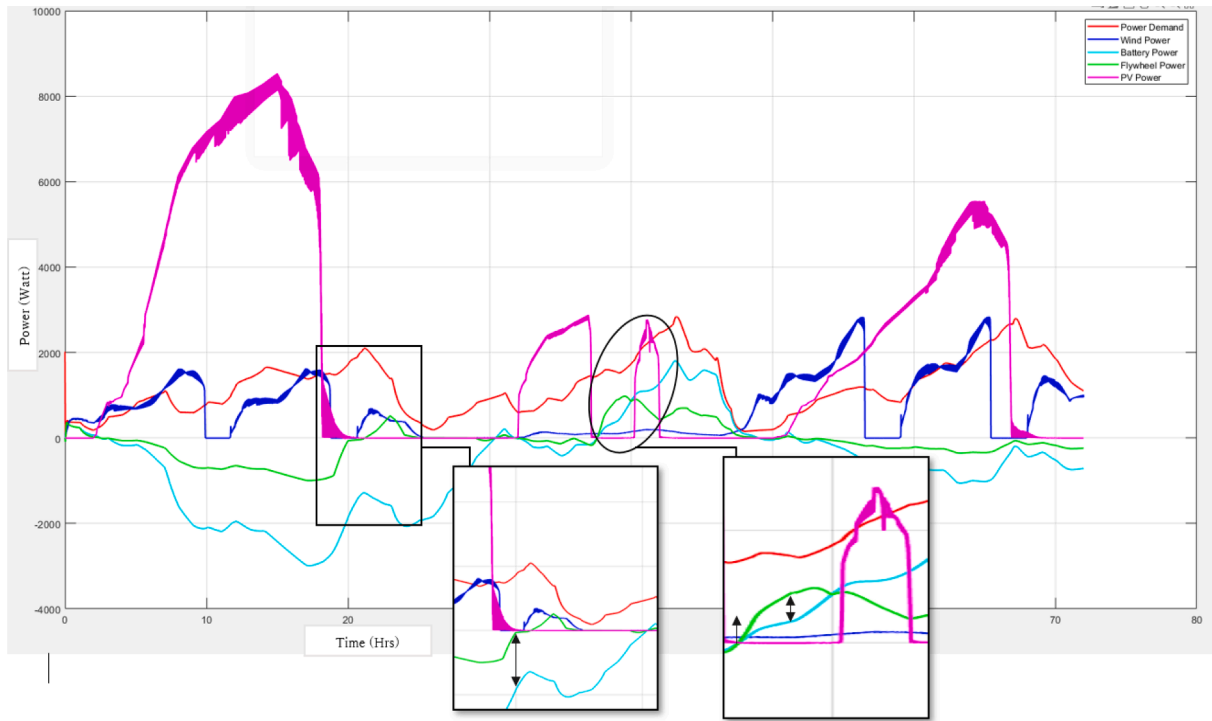


Fig. 30. Flywheel-Battery system response to load demand at low solar/wind power generation.

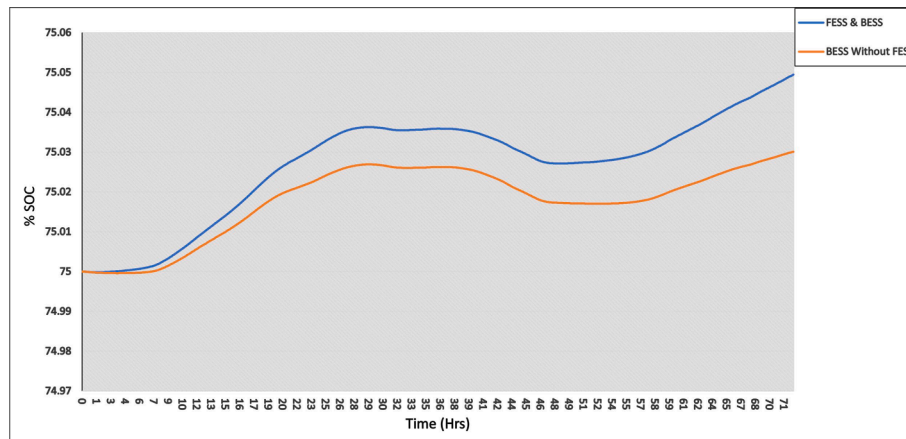


Fig. 31. State of charge (SOC) for the battery with and without the flywheel.

difference between capacity gain and capacity loss at the same battery SOC values due to flywheel-battery hybrid storage is 6330Wh. This provides valid evidence of the significant storage capacity achieved by the battery in hybrid energy storage combination in this research work.

Conclusions

Developing countries like Kenya and other sub-Saharan African countries require stand-alone microgrid systems to provide electricity in remote areas. However, a few considerations limit the choice of implementation of such microgrids: owing to the low-income status of residents in these regions, the cost of electricity must be affordable. Furthermore, most of these remote zones are entirely isolated from the grid system and consequently need highly reliable power supply systems. Moreover, the necessity to reduce the use of fossil fuel-based power generators, which emit greenhouse gases, further justifies the need to exploit effective and efficient storage technologies in

implementing clean energy microgrid systems.

This research introduced a comprehensive modeling approach to find the optimal size, configuration, and operation of a proposed PV-Wind-Battery-Flywheel HRES, to be used as an off-grid microgrid in remote areas in Kenya. To this aim, the optimal sizing model was founded on a multi-objective function optimization, considering both LCOE and reliability as the objective function. Using an applicable energy index, EISR, as a measurement tool in visualizing the relationship between the COE and the UME, a Pareto front was generated in which the optimal configuration of the system at the minimum cost and maximum reliability of the system can be found. In addition, a dynamic power control simulation model was developed in Matlab-Simulink in order to evaluate the hourly power supply demand of the HRES under different operating conditions.

The multi-objective optimization results for COE were realized at 0.519 USD/kWh depicting a suitable compromise for high-reliability levels and electricity affordability in order to satisfy a given local

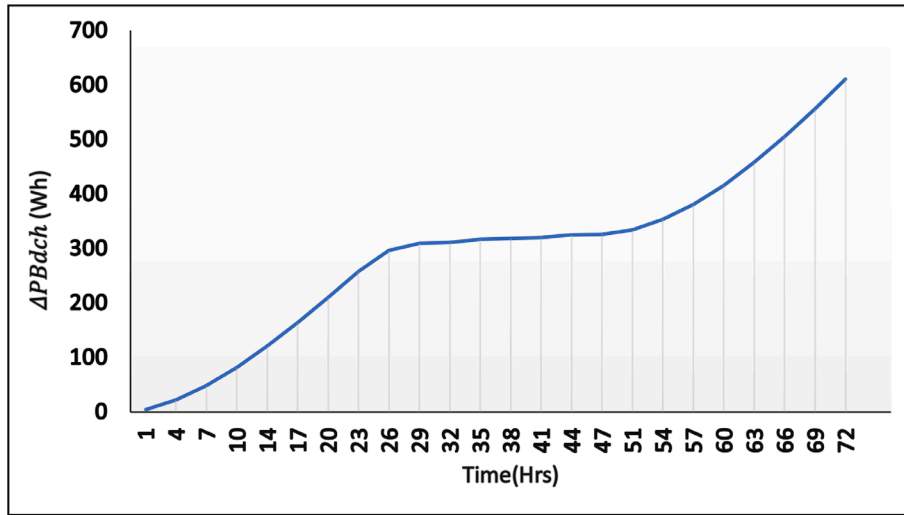


Fig. 32. Battery capacity gain with battery/flywheel storage system.

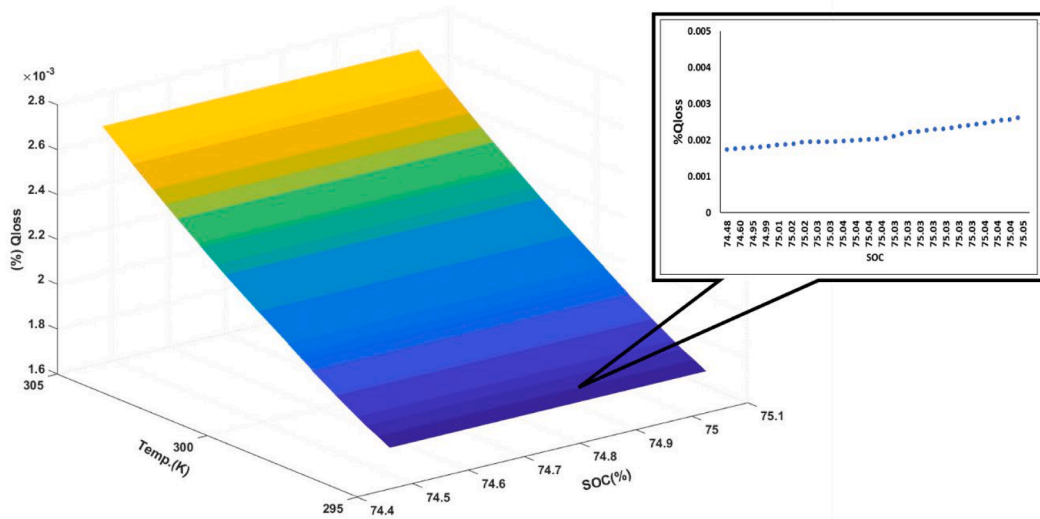


Fig. 33. Battery capacity loss with SOC and temperature.

demand requirement of 37.94MWh in Makueni county in Kenya, utilizing 26 solar PV panels, 3 wind turbines, and a HESS system made up of battery and flywheel to gain high reliability at affordable COE illustrates a commendable need to incline on renewable energy systems. The optimal storage capacity for the battery and flywheel, 22kWh and 4.7kWh, respectively, represent the need for storage to achieve energy reliability. The value of unmet energy, UME, was found at 0.00308, showing that HRES can supply electricity consistently.

The charge and discharge control for the battery and the flywheel provides a multi-faced benefit: the advantage in frequency response and the gain in battery capacity. The FESS demonstrates a fast frequency response in the event of a sudden halt in power generation by the PV and wind turbine systems. Such conditions, where no radiation and wind speeds are very low, lead to an unexpected overreliance on the storage system. As demonstrated and discussed in this study, the flywheel storage system provides a fast dynamic response and high energy density necessary to avoid possible power outages. A comparison and analysis of battery storage capacity behavior both with and without hybrid storage in the HRES system showed a significant increment in capacity when the BESS-FESS storage system is assumed.

CRediT authorship contribution statement

Alphonse Ngila Mulumba: Conceptualization, Methodology, Software, Investigation, Writing – original draft. **Hooman Farzaneh:** Conceptualization, Methodology, Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix

Tables A1-A4

Table A1
Solar panel parameters [41].

Parameters	Value
Rated power	330 W
Maximum voltage at STC, $V_{m(STC)}$	41.9 V
Open circuit voltage, V_{oc}	50.9 V
Open circuit voltage at STC, $V_{oc(STC)}$	37.3 V
Maximum Current at STC, $I_{m(STC)}$	7.88A
Short-circuit current, I_{sc}	8.47A
Short-circuit current at STC, $I_{sc(STC)}$	8.78A
Current temperature coefficient, $\mu_{I_{sc}}$	-173.1 mV/°C
Voltage temperature coefficient, $\mu_{V_{oc}}$	3.6 mA/°C
PV empirical adjustment parameter, σ_{oc}	-0.04
Thermal Voltage, V_{th}	25.7 mV
Cell Temperature at STC, $T_c(STC)$	25 °C
Irradiance at STC, $G_{(STC)}$	1000 W/m ²

Table A2
Flywheel parameters.

Parameters	value
Flywheel poles	8
Inner radius	0.135 m
Outer radius	0.15 m
height	0.6 m
Rotor mass	20 kg
Mass density	760 kg/m ³
Minimum rotation speed	1000 rpm
Maximum rotation speed	9000 rpm

Table A3
Wind turbine parameters.

Parameters	Value
Rated power	1 kW
Rated voltage	110 V
Startup wind speed	1.5 m/s
Rated wind speed	10 m/s
Maximum wind speed	50 m/s
Rated rotation speed	300U/min
Fan blade quantity	3
Rotor blades diameter	2 m
Tip speed	31 m/s

Table A4
Battery parameters.

Parameters	value
Type	Lithium-ion
Fully charge voltage	14 V
Capacity	3000 Ah
Charging current	2A-10A
Charging Voltage	14.4-14.9 V
Float voltage	13.5-13.8 V
I_{rate}	6C

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